

Problem H3.1 [10 points]

- Write a function `my_factorial` that recursively evaluates the factorial of an arbitrary integer $n \geq 0$. Remember that $0! = 1$ and $n! = n \cdot (n-1)!$ **(3pt)**
- Write a function that calculates the Taylor expansion

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots,$$

that also uses your previously defined function `my_factorial`. The function should take the number of terms in the expansion as a keyword argument (i.e. with a default value), and the angle x in radians as an argument. Since you know the number of terms, you should use a for-loop to calculate the Taylor expansion. **(4pt)**

- Using $x = \pi/2$, compare your function to `numpy.sin()`. Print the results alongside the name of the function used, using the f-string to show the numbers with at least 5 decimal places. **(1pt)**
- Use `%timeit` to show which function is faster. **(1pt)**

Note: The remaining 1 point is earned by writing concise, elegant and efficient code.